

TimeCart — System Documentation

Professional Technical Report

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2026-09-14

v0.1.0

Table of Contents

- 1. Introduction ...**
- 2. Database Design ...**
- 3. Key Workflows ...**

1. Introduction

TimeCart ("Where Time Meets Style") is a premium watch e-commerce storefront. From the project README and source, it is built with Next.js 16 (App Router), Supabase (Postgres / Auth / Storage), Prisma 7, Tailwind CSS 4, and Recharts. It supports customer browsing, cart, checkout, order tracking and reviews, plus an admin area for products, coupons, customers and orders.

- Frontend and API: Next.js 16 App Router (evidence: package.json, src/app).
- Database engine: relational Postgres via Supabase (evidence: prisma/schema.prisma, datasource db).
- Authentication: Supabase Auth with custom middleware (evidence: src/middleware.ts, src/lib/session.ts).
- Email delivery: Resend (evidence: package.json dependency + usage).

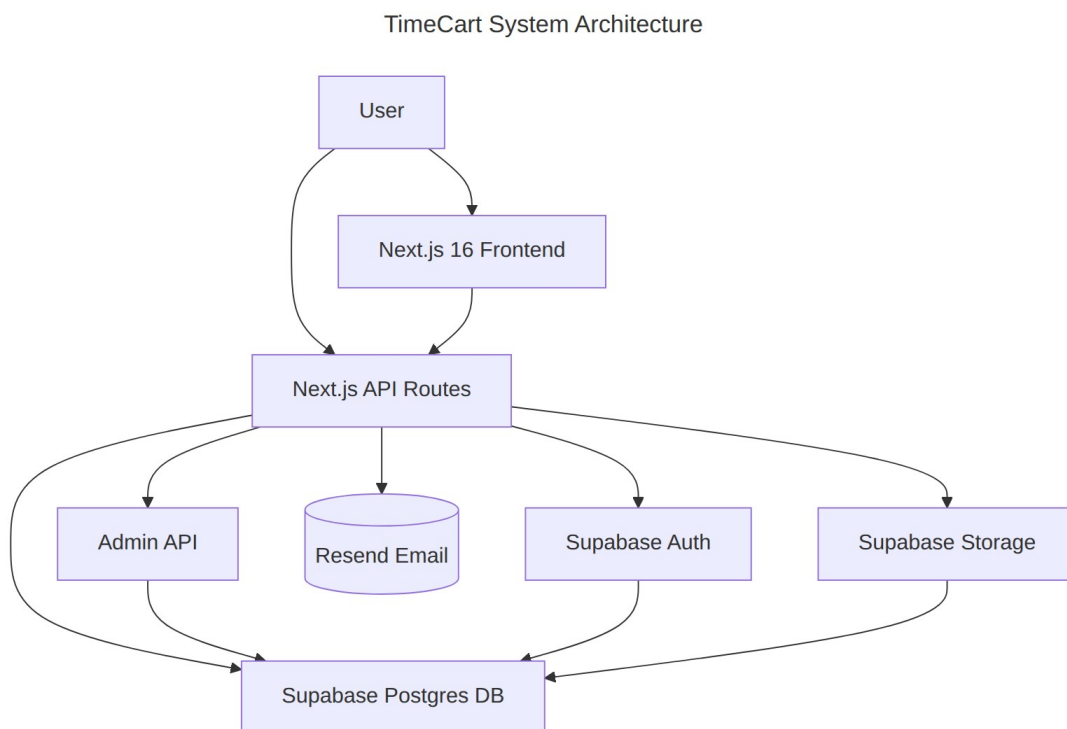


Figure 1.1 — System architecture (from source structure and configuration).

2. Database Design

The relational schema is defined in `prisma/schema.prisma` as a Postgres database with 23 models (e.g. User, Profile, Product, Order, OrderItem, Payment, Review, Coupon) and 10 enums (UserRole, OrderStatus, PaymentMethod, ...).

Table 2.1 — Core tables detected in the Prisma schema.

Table	Key fields / purpose
User	auth accounts; role enum CUSTOMER ADMIN

Profile	1:1 with User; name, contact details
Product	catalog item; brand and category relations
ProductVariant	size/color stock per product
Order	customer order; status enum lifecycle
OrderItem	line items per order; product relation
Payment	1:1 with Order; method + status
Review	product reviews by user
Coupon + CouponUsage	discounts and their redemption tracking

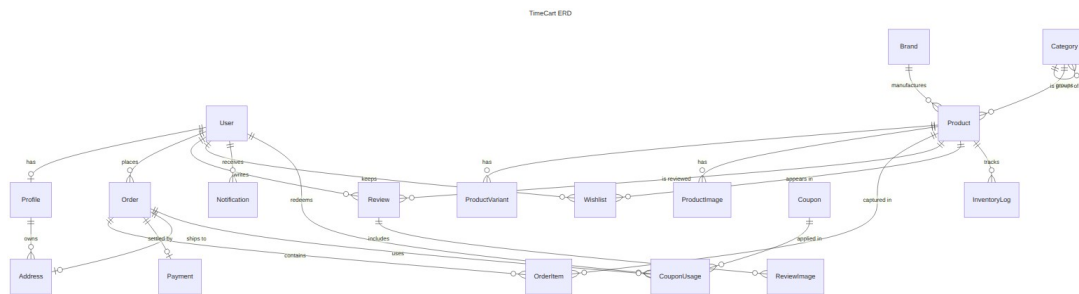


Figure 2.1 — Entity Relationship Diagram (generated from prisma/schema.prisma relations).

3. Key Workflows

Authentication flow: the user signs in to the Next.js frontend, the API route delegates to Supabase Auth and a session is stored; subsequent requests go through middleware and read session data (evidence: `src/app/auth/callback`, `src/lib/session.ts`, `src/middleware.ts`).

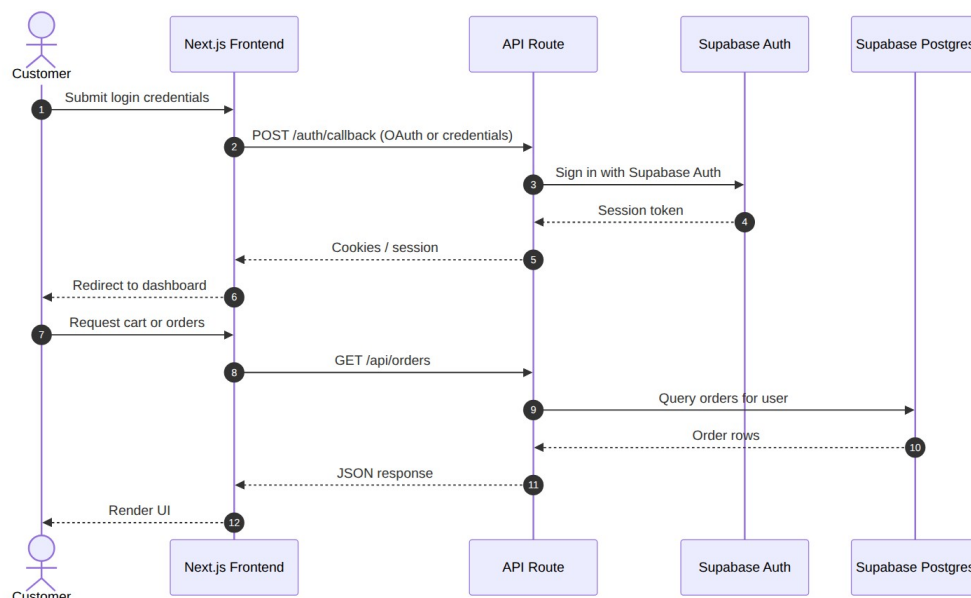


Figure 3.1 — Login / session sequence (from auth callback route and lib code).

Public API surface observed (`src/app/api`): products, search, reviews, coupons/validate, orders, track-order, wishlist-related, newsletter, contact, config, upload, profile, and an

/api/admin/* set (customers, orders, products, reviews, brands, coupons, categories, contact).